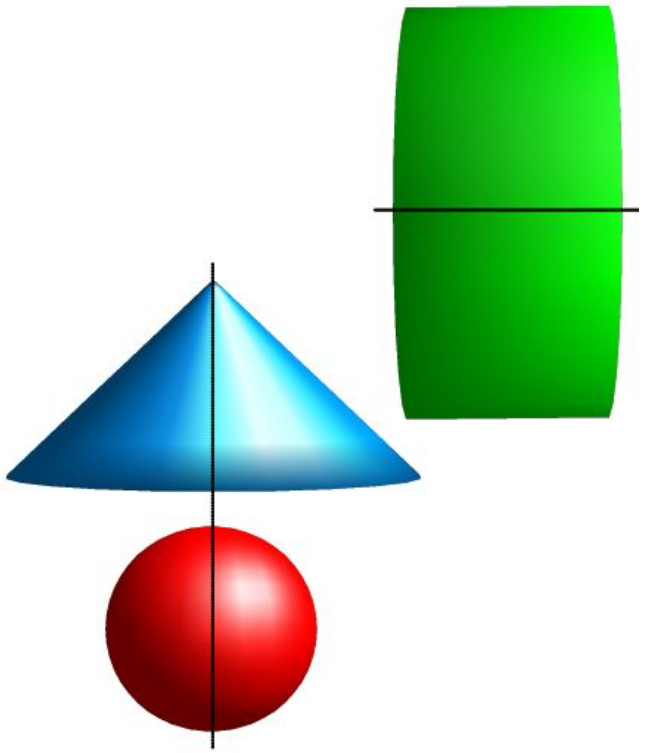
Archimedes used the Law of the Lever to compare volumes of a cylinder, cone and sphere.

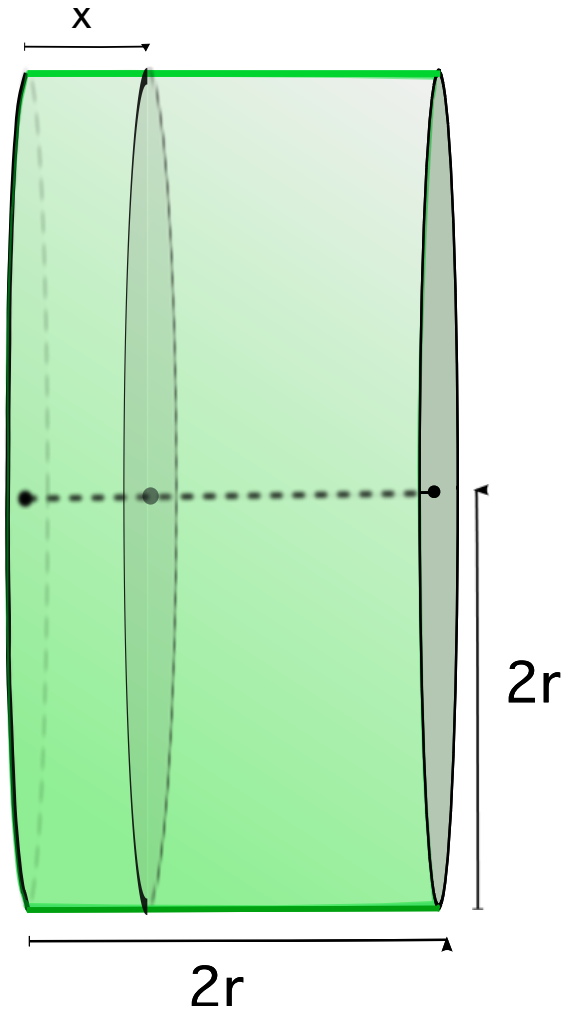


Archimedes knew the volumes of cylinders and cones. and areas of their circular cross sections. He used these to determine the volume of the sphere.

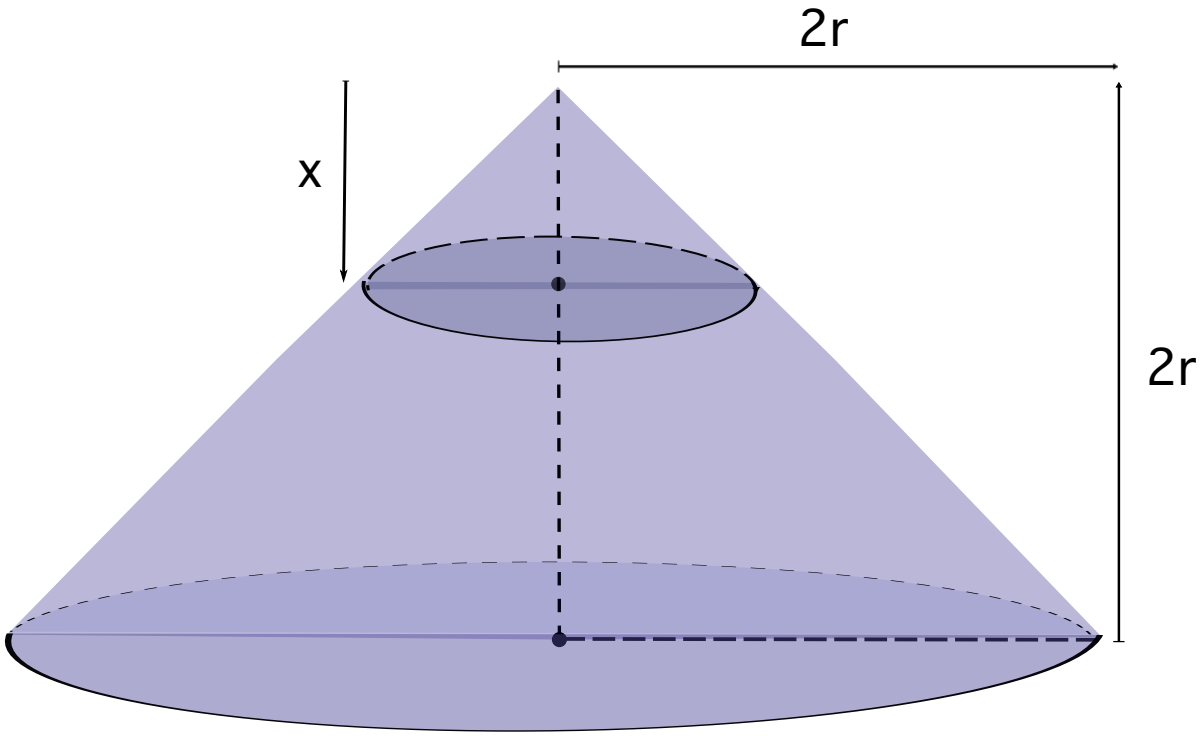
Cylinder: Area of x-level Cross Section

Circular cross-section at level x: $4\pi r^2$

[Volume of cylinder = $C_y = 8\pi r^3$]



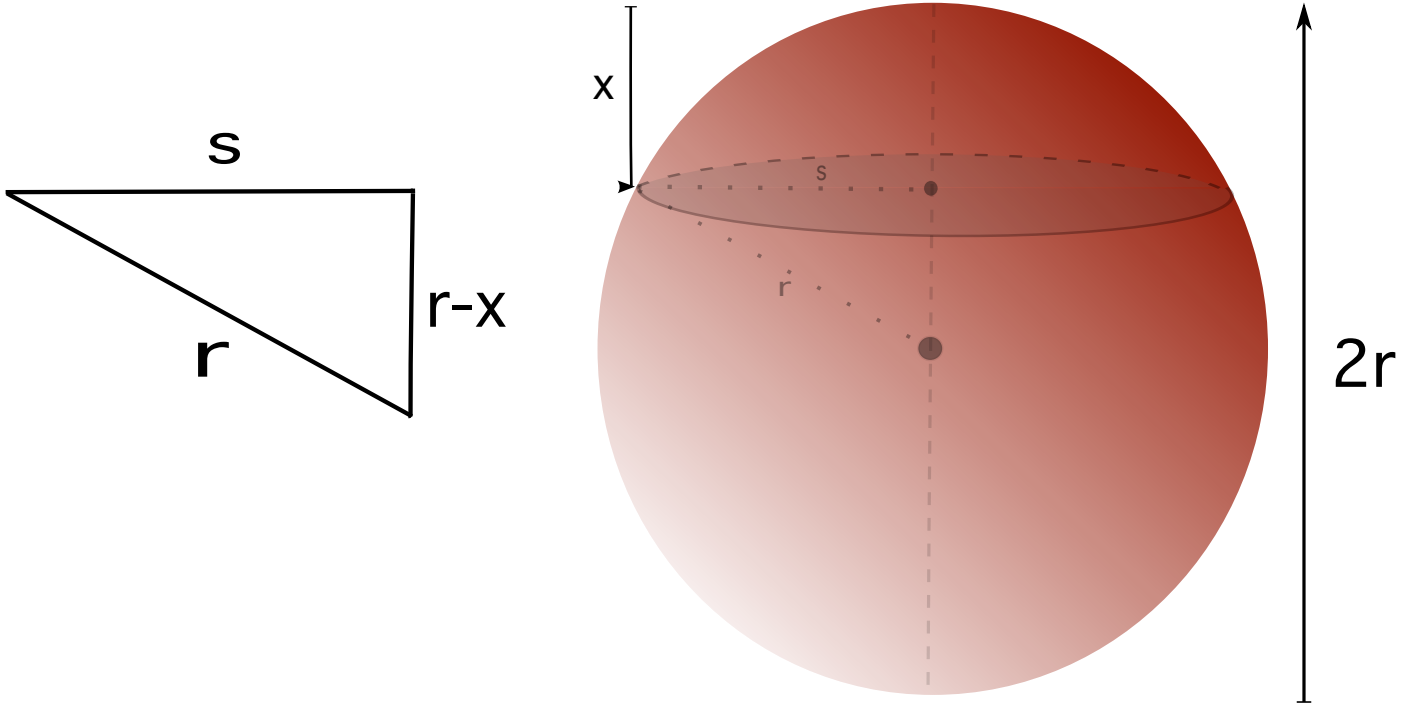
Cone: Area of x-level Cross Section



Circular cross-section at level x: πx^2

[Volume of cone = $C_o = \frac{8}{3}\pi r^3$]

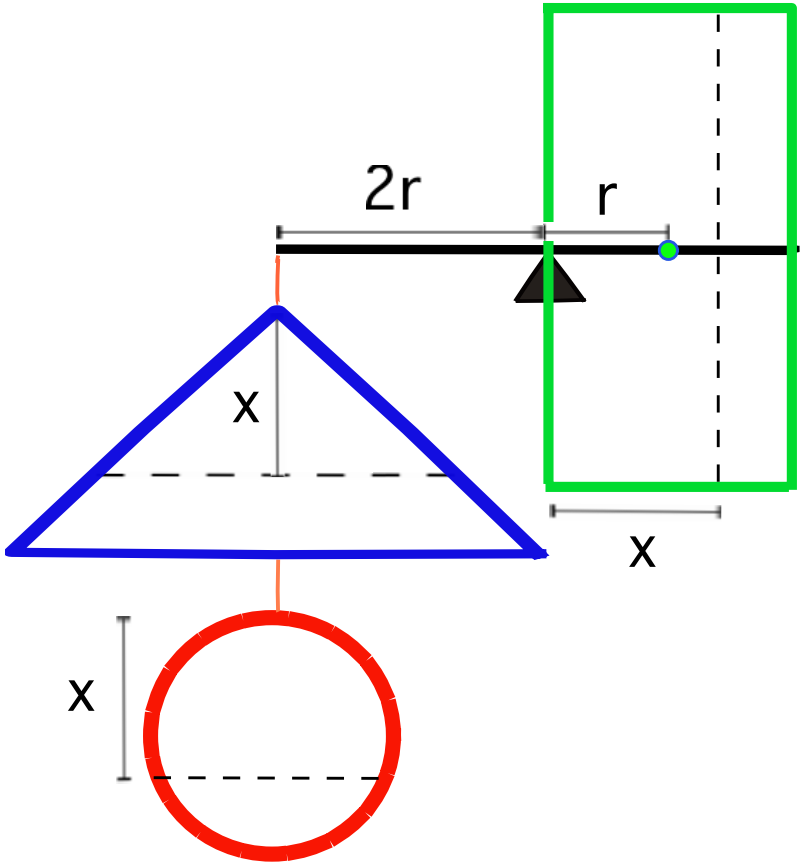
Sphere: Area of x-level Cross Section



Circular cross-section at level x**:** $\pi s^2 = 2\pi r x - \pi x^2$

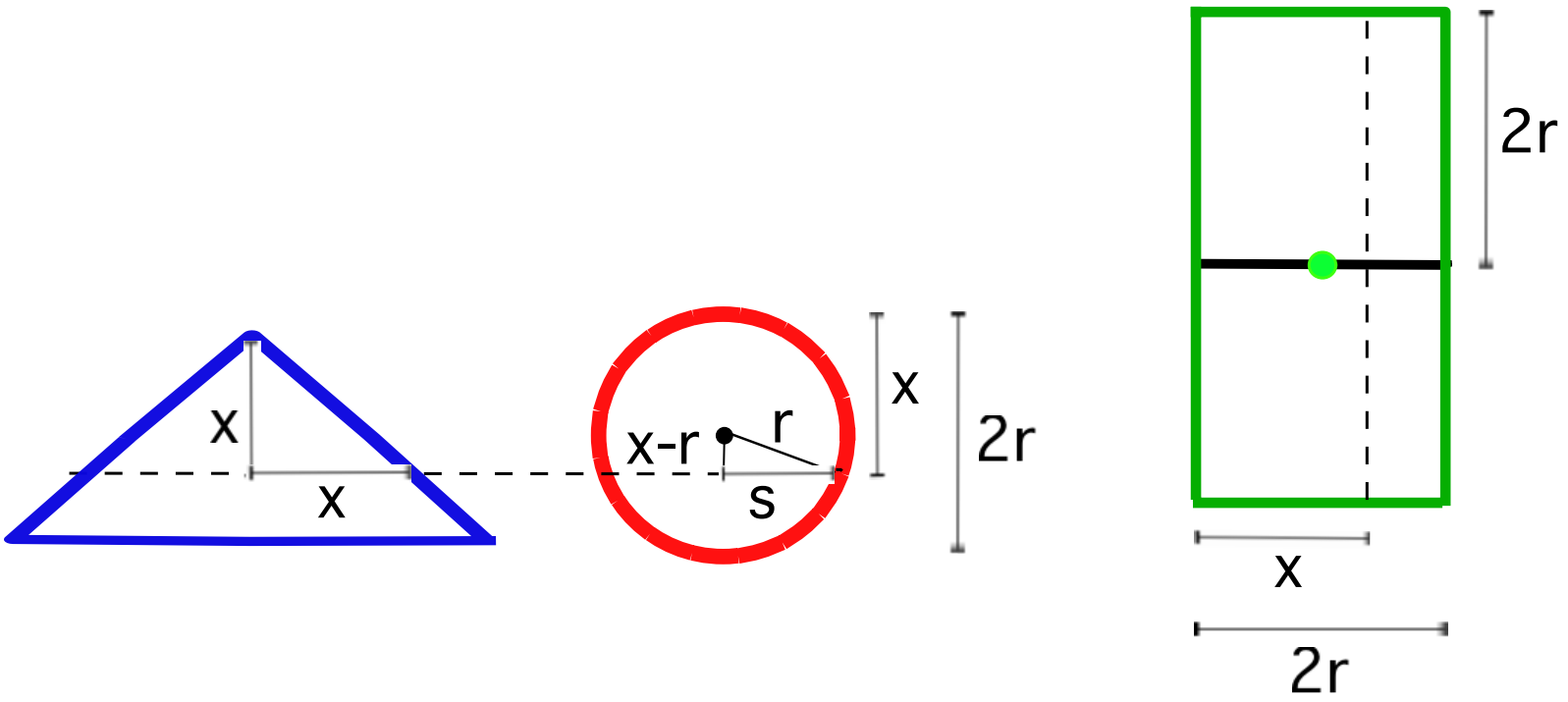
[Volume of sphere = $S = ???$]

Method (Step 1): Three Solids in Perfect Balance



We will see that the x-level cross sections balance. (Assuming uniform density.)

Method (Step 2): The cross-section (CS) areas balance.



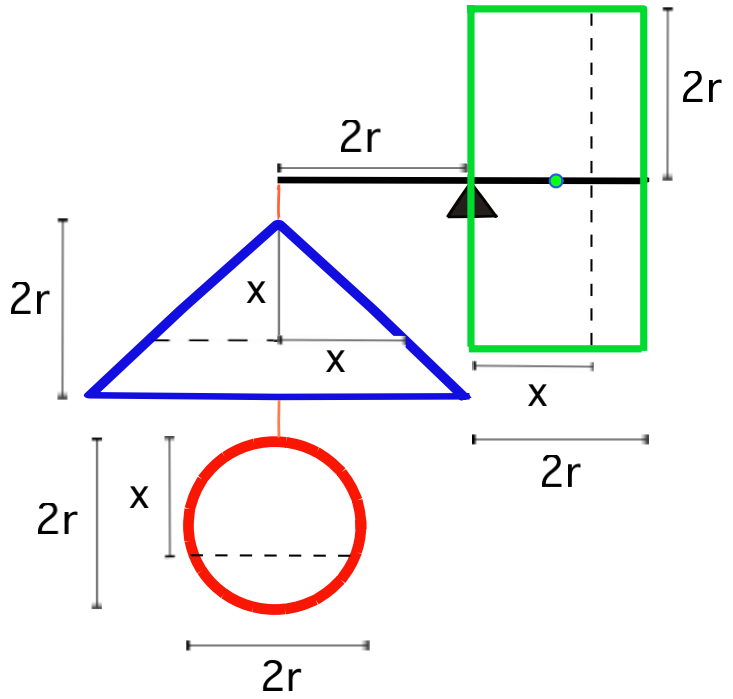
Cone CS: $+\pi x^2;$

Sphere CS: $\pi s^2 = 2\pi r x - \pi x^2;$

(Cone CS + Sphere CS) = $2\pi r x$ Cylinder CS: $4\pi r^2$

2r · (Cone CS + Sphere CS) = $4\pi r^2 x$ = Cylinder CS · x **QED**

Method (Step 3): The Derivation



By previous slide: Each x-section of cylinder balances the combined x-sections of the hanging cone and sphere. Therefore, **the volumes balance**: $2r \cdot (C_o + S) = r \cdot C_y$, so

$$C_o = \frac{8}{3}\pi r^3, \quad C_y = 8\pi r^3 \quad \Longrightarrow \quad S = \frac{4}{3}\pi r^3$$

Using modern terms, we would express this as:

$$2r(C_o + S) =$$

$$\int_0^{2r} (2r)(2\pi rx)dx = 8\pi r^4 = \int_0^{2r} (x)(4\pi r^2)dx$$

$$= rC_y$$

Archimedes did this nearly 2 millenia before Newton and Leibniz.